WEILL CORNELL MEDICINE · ZELEZNIK LABORATORY

Inside SeNMo

Multi-omics survival modeling on the TCGA pan-cancer atlas — from first-principles data exploration to a reproduced, HPC-scale preprocessing pipeline.

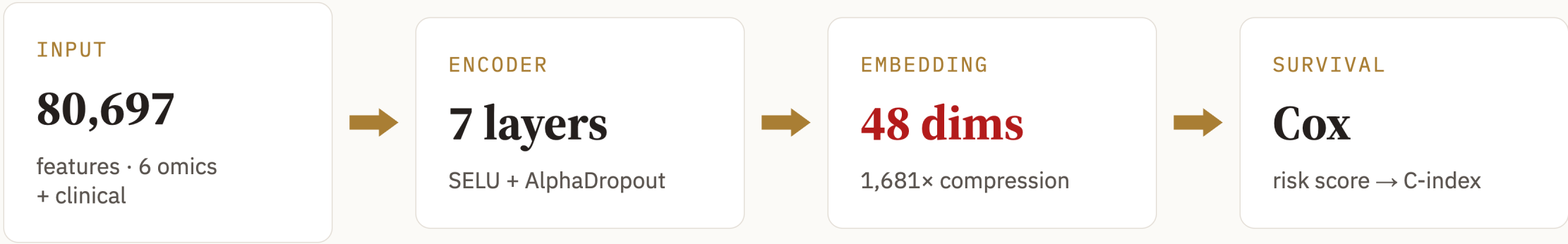
Samantha Fernandez Perez
Post-graduate Research Intern · Summer 2026

Supervised by Dr. Oana
Mentorship: Nan

THE MODEL

One model, one bottleneck

SeNMo is a self-normalizing multi-omics network: it compresses every patient's molecular profile through a narrow encoder and predicts overall survival with a Cox objective.



Self-normalizing activations (SELU + Alpha-Dropout) hold the signal stable through depth without batch-norm — which matters because the six omics live on wildly different numeric scales.

THE THESIS UNDER TEST

Age and sex should be *first-class variables* in a survival model — not one diluted column among 80,697 features.

Ovarian cancer (anatomically all-female) is the entry point; the broader claim is that demographic biology belongs in the representation itself, not averaged away by a sex-pooled cohort.

THE ARC

Seven weeks — and then the cluster

WEEK 1

Foundations

Architecture & the hypothesis

WEEK 2

TCGA-OV

Cohort exploration & survival stats

WEEK 3

ML bridge

Every concept mapped to code

WEEK 4

Replication

Reproduce Fig 7 + probe the embedding

WEEK 5

Cross-cancer

BRCA, LUAD & the female pipeline

WEEK 6

The pivot

Demographics as first-class

WEEK 7

To the server

HPC access on Cayuga

WEEK 8 +

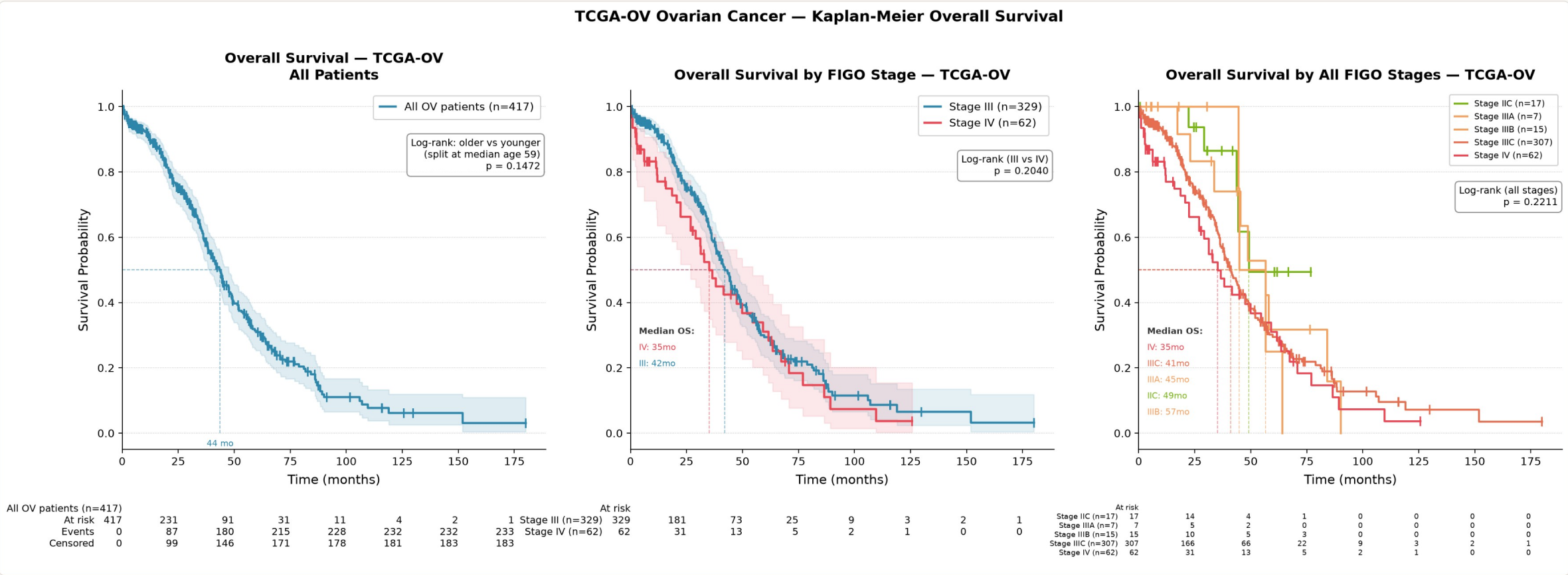
On the cluster

Reproducing preprocessing at scale

WEEKS 1–2 · FOUNDATIONS

Reading the data before the model

tcga_ov_exploration.py

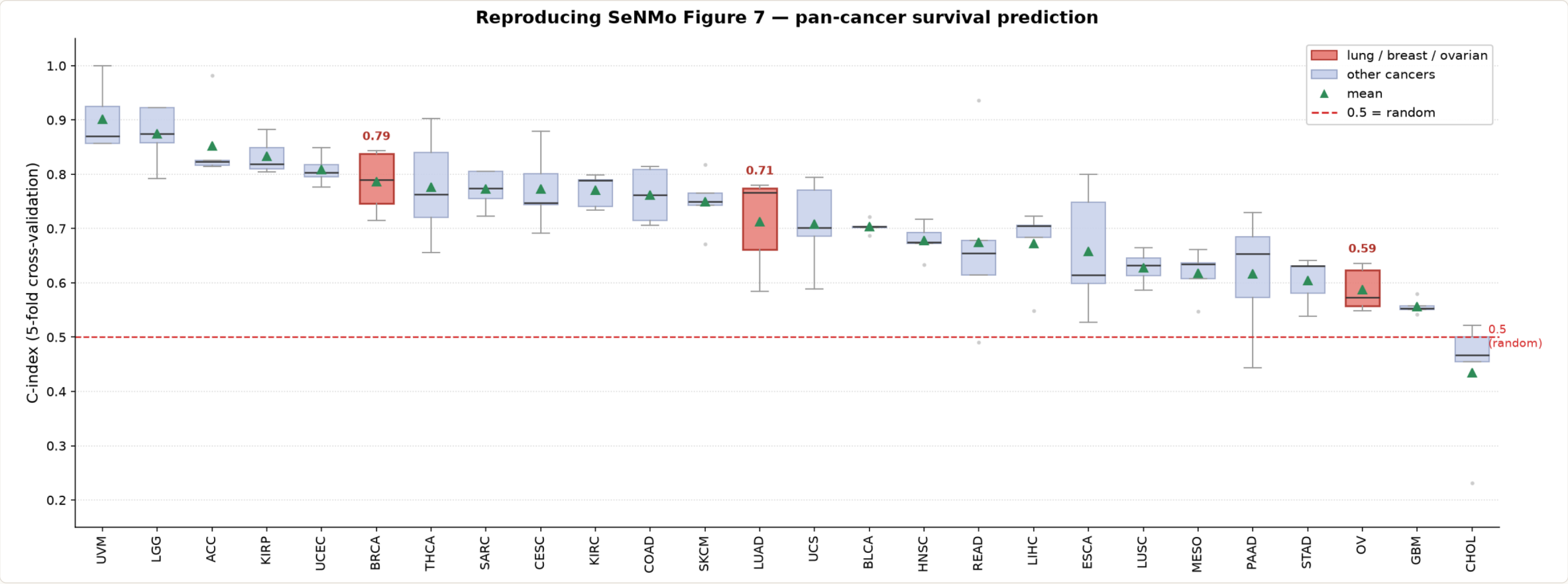


Survival analysis from first principles — Kaplan–Meier, Cox, concordance index. TCGA-OV overall survival (n=417): overall, by Stage III vs IV, and across all FIGO stages, with at-risk tables and log-rank tests. Median OS ≈ 44 mo; a clinical-only Cox baseline reaches **C-index 0.626** — the floor the embedding must beat.

WEEK 4 · THE MILESTONE

Reproducing the paper's Figure 7

reproduce_figure7.py



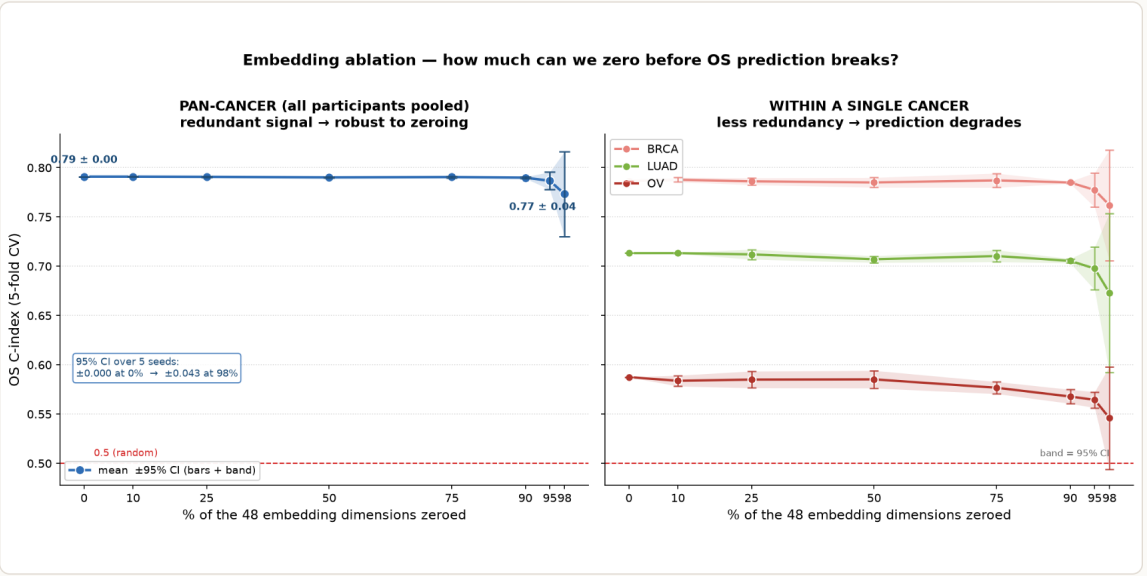
Per-cancer survival concordance from the frozen embedding (5-fold CV). Project targets highlighted: **BRCA 0.79, LUAD 0.71, OV 0.59**. Point estimates match the published figure — ovarian being genuinely hard is the reportable finding, not a bug.

WEEK 4 · PROBING

What does the 48-dim embedding encode?

ABLATION — REDUNDANCY

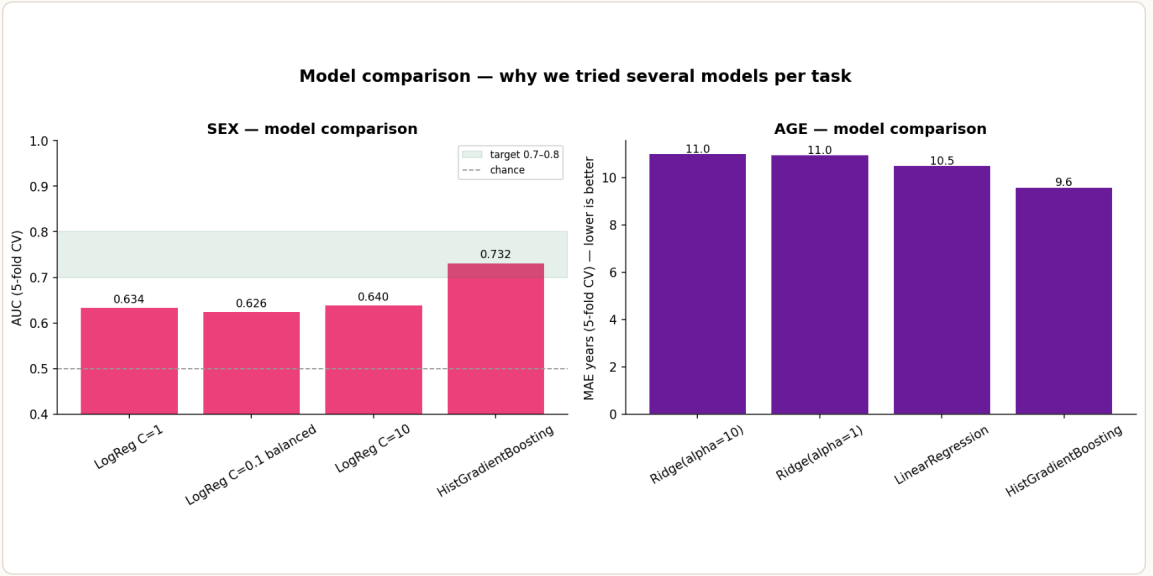
ablation_zeroing.py



Pan-cancer C-index holds at 0.79 even after zeroing 90% of dims; within one cancer it decays — the easy "which-cancer" signal is stored many times over.

AGE & SEX — MODEL COMPARISON

predict_age_sex.py

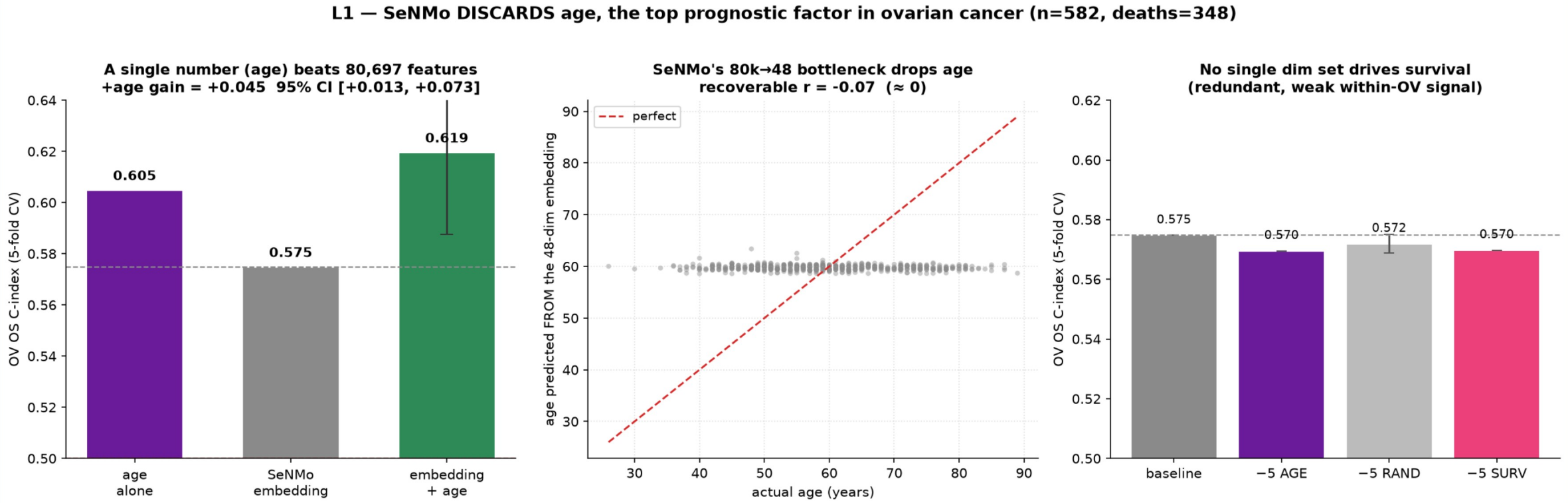


Demographics are recoverable only non-linearly: sex AUC 0.732 and age MAE 9.6 yr with gradient boosting, while linear probes plateau near 0.63.

WEEK 6 · THE DECISIVE RESULT

SeNMo discards age

l1_ov_age_steering.py



In ovarian cancer (n=582), **age alone (0.605) beats the full 48-dim embedding (0.575)**; adding age back gives +0.045. Age is nearly unrecoverable from the embedding (r ≈ −0.07) — the 80k→48 bottleneck squeezes it out.

THE CONVERGED NARRATIVE

Three lines of evidence, one conclusion

L1

Age is dropped

The embedding loses age across nearly every cancer; recovering it adds +0.045.

L2

Recovery is additive

Conditioning (FiLM) doesn't beat concatenation — a head-level patch is trivial. The real fix is upstream.

LUAD

Filtering buys nothing

Female-only (0.675) ≈ mixed (0.679) on the same held-out women. Male data is neither noise nor much signal.

THE CONCLUSION

The lever is *demographic recovery* — retrain the encoder with age & sex first-class, not sex-filtering.

WEEK 7

To the server

Every result so far was a frozen-embedding probe — possible on a laptop precisely because the encoder itself cannot be trained there. That needs a GPU cluster. Week 7 is the transition from analysis to infrastructure.

NOW

Connect

SSH to Cayuga · build the env

NEXT

Download

Slurm job pulls TCGA omics

THEN

Preprocess

replicate the pipeline at scale

GOAL

Train

retrain the encoder

Cornell CAC "Cayuga" cluster · Slurm scheduler · Xena ETL for five omics + phenotype + survival · GDC MAFs for mutations.

CHAPTER 8 · ON THE CLUSTER

Reproducing the preprocessing at pan-cancer scale

The work after the written portfolio: rebuilding SeNMo's six-omic pipeline across all 33 TCGA cohorts on Cayuga — faithfully, and fast enough to actually run.

RESULTS

81,240 features — within 0.7% of the authors

The assembled matrix is **11,127 patients × 81,240 features**. Two omics match to the exact feature; every omic is within 2.4%.

Omic	Authors	Ours	Δ%
Clinical	4	4	0.0
Protein (RPPA)	472	472	0.0
miRNA	1,730	1,746	+0.9
Gene expression	8,794	8,582	−2.4
DNA methylation	52,396	53,387	+1.9
DNA mutation	17,301	17,078	−1.3
6-MODAL TOTAL	80,697	81,240	+0.7

Residual gaps live only in the **tunable, data-dependent filters** — clinical and protein, which have no such filter, match exactly.

ENGINEERING

Making it run — and making it correct

PARALLELISM

Slurm job arrays

All 33 cohorts processed in parallel, each picking its cohort from the array index — a multi-day serial job down to hours.

MEMORY

Parquet checkpoints

Wide CSV writes on ~130,000-column matrices ran out of memory and died. Parquet writes the same data in seconds.

CORRECTNESS

A 6,197-sample bug

A survival-label join silently duplicated 6,197 samples. Diagnosed, fixed by holding labels raw — duplicates → 0.

The statistical operations are the authors' code verbatim; only the execution harness — array indexing, Parquet, absolute paths — is adapted to run headless at scale.

VALIDATION

Checking against the authors' patients

The authors released only their **combined** embedding set, never their train/test split — so a patient-by-patient train match is impossible. The honest check is **containment**: is every patient I kept one the authors also had?

It holds. The patients I'm missing are precisely their held-out **test split** — expected, since I built only the training side.

Authors (train + test)	**10,938**
Mine (training matrix)	**9,248**
Shared patients	**9,151**

Authors-only 1,787 (their test split) · mine-only 97 (~1%, under review) · Jaccard 0.83.

OUTCOMES

What the work built

Skills

- Survival analysis — Cox PH, C-index, Kaplan–Meier
- Multi-omics & TCGA / UCSC Xena data
- HPC — Slurm job arrays, batch workflows
- Reproducible, auditable pipelines
- Python · pandas · PyTorch · scikit-survival

Delivered

- Reproduced the paper's Figure 7
- Reproduced full preprocessing within 0.7%
- Found SeNMo systematically drops age
- Caught & fixed a 6,197-sample bug
- Sharpened the thesis with evidence

SEVEN WEEKS — AND THE CLUSTER

From a Kaplan–Meier curve to a pan-cancer pipeline.

An apprenticeship, a reproduction, a genuine finding, and the infrastructure to build on — a defensible, evidence-driven redirection of a foundation-model project.

Samantha Fernandez Perez

Zeleznik Laboratory · Weill Cornell Medicine · 2026